Intelligence Compromised

China's Embedded Access to IARPA and U.S. Defense Research

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Foreword

The *Intelligence Compromised* report is a pioneering open-source magnum opus. With unmatched rigor and clarity, it exposes the deep entanglements that China's defense research and industrial base have forged with U.S. intelligence research programs—particularly those funded by the Intelligence Advanced Research Projects (IARPA)—and with the American universities that conduct them. These linkages constitute grave national security threats to the United States and its Free World allies.

Such engagements never occurred with the Soviet Union during the first Cold War. They must not be tolerated in this new Cold War with our greatest strategic adversary, the People's Republic of China. The report lays bare the stark asymmetry: while Beijing wages unrestricted warfare across every domain—military-civil fusion, talent programs, open-source exploitation, and systematic technology transfer—the United States has too often remained on a peacetime footing, subsidizing the very capabilities that now threaten our technological edge and the common defense.

When serving as United States Ambassador to Fiji, Kiribati, Nauru, Tuvalu and Tonga from 2019-2021, I witnessed firsthand the Chinese Communist Party's malign influence across the Indo-Pacific - through aggressive diplomacy, infrastructure incursions, and attempts to undermine U.S. alliances in the Pacific Islands. This experience mirrors the threats detailed in this report: Beijing's waging of unrestricted warfare through the relentless pursuit of technological and strategic dominance, whether by co-opting island nations or infiltrating American research institutions. The exploitation of IARPA programs is but one front in this broader assault, demanding the same vigilance my Free World counterparts and I applied in countering PRC expansionism abroad.

The authors have performed an invaluable public service. Through meticulous case studies, they document how IARPA-funded breakthroughs in biometrics, quantum sensing, AI-driven surveillance, and cognitive analytics have been openly analyzed, replicated, and adapted by institutions tied to the PLA, the Ministry of Public Security, and the Ministry of State Security. This is not academic curiosity; it is strategic exploitation, and it is unconscionable.

On a whole-of-society and whole-of-government basis, the United States and its partners must now conduct a deep, unflinching self-assessment. The time has come for deliberate de-risking—and, in critical sectors, decisive decoupling—from entities that serve the Chinese Communist Party's military modernization. This report provides a clear roadmap in one area where that urgent work must begin. America's technological superiority was never meant to be a gift to our adversaries. It is time to reclaim it.

Joseph Cella
Former United States Ambassador Fiji, Kiribati, Nauru, Tuvalu, Tonga (2019-2021)

Executive Summary[1,2]

This report exposes a persistent and strategic pattern of exploitation by China's defense research and industrial base[3] targeting U.S. intelligence research programs, particularly those funded by the Intelligence Advanced Research Projects Activity (IARPA). Through systematic analysis, collaboration, and dual-funding arrangements, institutions tied to the Chinese Communist Party (CCP), People's Liberation Army (PLA), and Ministry of Public Security (MPS) have accessed, sought to replicate, reverse-engineer, and adapt IARPA taxpayer-funded projects, many of which have clear national security implications. All while China's internal surveillance system, anchored by MPS projects like Skynet, Sharp Eyes, and Safe City, has evolved into a nationwide biometric dragnet enabling real-time tracking of both citizens and foreign operatives.

Despite IARPA's Research and Technology Protection (RTP) process being hailed by the Office of the Director of National Intelligence (ODNI) as far back as 2014, and other federal authorities as a model of security best practices, safeguards ultimately failed to prevent adversarial exploitation. China's defense research and industrial base have openly dissected IARPA's technical objectives, acquisition models, and national security value chains to inform China's own R&D efforts in surveillance, AI, behavioral analytics, and cognitive warfare.

Chinese police and military universities, including the Chinese People's Public Security University, the PLA Information Engineering University, and the PLA's National University of Defense Technology (NUDT), have published detailed assessments of IARPA's behavioral analytics, anticipatory intelligence frameworks, and project structuring. These assessments are operational analyses conducted to identify, extract, and adapt U.S. innovations for deployment in Chinese state security operations. This activity fits squarely within China's broader military-civil fusion strategy (MCF, 军事民用融合), which is a Chinese national strategy that seeks to eliminate barriers between the civilian and military sectors, enabling the PLA to directly access and apply cutting-edge technologies developed in academia, industry, and abroad. Under MCF, civilian institutions and companies are legally obligated to share innovations with the PLA to accelerate China's military modernization.

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1 The views expressed in this academic research paper are those of the authors and do not necessarily reflect the official policy or position of the U.S. Government, the House of Representatives, or the Select Committee on China.

2 Mention of individuals, companies, research institutions, or other entities in this volume does not imply a violation of any law or international agreement and should not be construed as such. The authors of this study make no claims that any of the named US entities engaged in any illegal activity.

3 China's defense research and industrial base: People's Liberation Army organs, State-Owned Enterprise defense conglomerates, Seven Sons of National Defense universities, Seven Sons or Ordinance (Arms) Industry universities, State Administration for Science, Technology, and Industry for National Defense co-administered universities, national defense-designated laboratories, and the intelligence and public security apparatus.

Case studies documented in this report reveal PRC-affiliated researchers participating in active IARPA-funded projects across a range of sensitive disciplines, projects designed specifically to advance the capabilities of the U.S. Intelligence Community (USIC). These engagements, many of which occurred in the open through dual affiliations, joint publications, or subcontracting, reveal a critical breakdown in America's research security posture and due diligence efforts. This raises serious concerns about the effectiveness of IARPA's RTP program in conducting research security, due diligence vetting, and post-award compliance and monitoring.

IARPA's own mission statement makes its national security orientation unmistakably clear: "to push the boundaries of science to develop solutions that empower the IC to do its work better and more efficiently for national security." While IARPA itself does not deploy technologies operationally, it explicitly states that it "facilitates the transition of research results to [its] IC customers for operational application."[4] In other words, the fundamental purpose of IARPA-funded research is to equip the USIC with strategic advantage through cutting-edge scientific innovation. Given this mission, it raises a critical and urgent question: **Why are researchers funded by IARPA awards not explicitly prohibited from conducting joint research or collaborating with entities tied to China's defense research and industrial base?**

The USIC must prohibit any collaboration, direct or indirect, between intelligence research programs and China's defense research and industrial base. This includes joint publications, subcontracting, co-funding, or participation by any entity affiliated with China's defense research and industrial base. Absent immediate policy action, rigorous enforcement mechanisms, enhanced analytic due diligence vetting, and a reassessment of failed protection frameworks, the failure to close these channels not only perpetuates the erosion of U.S. technological superiority but also directly subsidizes the advancement of the CCP's military modernization goals.

Furthermore, allowing IARPA-funded researchers to simultaneously conduct research with adversarial entities, even indirectly through publications, joint labs, or shared pre-published research results, fundamentally undermines the purpose of the U.S. taxpayer-funded intelligence research enterprise. It reflects a policy gap that requires immediate correction.

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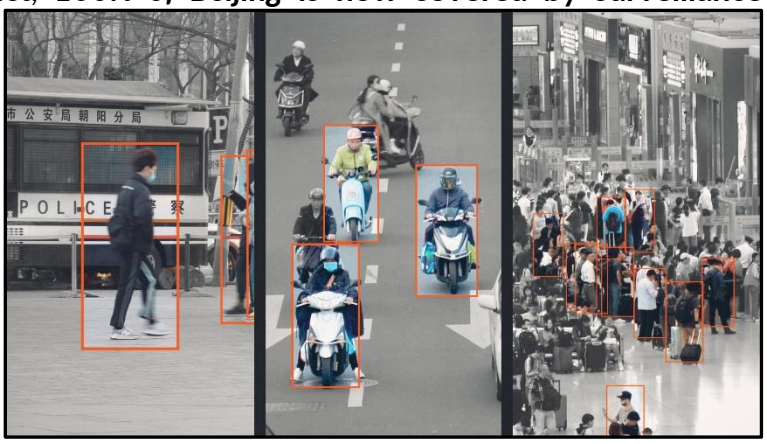
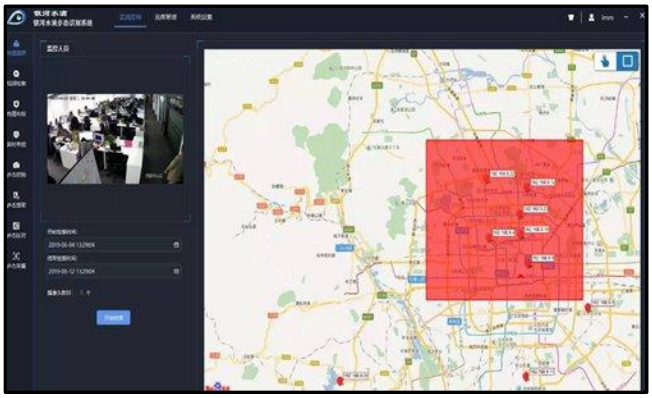
4 See https://www.iarpa.gov/who-we-are/about-us, Date accessed October 7, 2025

Total Surveillance: How China's Public Security System Tracks Citizens, Foreigners, and Spies

China's Ministry of Public Security (MPS) has built an extensive high-tech surveillance network across the country in recent years. The MPS oversees programs such as **"Skynet", "Sharp Eyes" (Xueliang), and "Safe City",** which blanket urban and rural areas with millions of CCTV cameras enhanced by artificial intelligence. In fact, ***100% of Beijing* is now covered by surveillance cameras**, and nationwide the plan was to deploy roughly 450 million cameras by 2020. The goal, first articulated by the MPS in 2015, was an "omnipresent, completely connected, always on and fully controllable" video surveillance network leveraging AI-driven facial recognition to catch lawbreakers in real time. By combining these pervasive cameras with advanced analytics, public security authorities can automatically identify individuals, track their movements, and even publicly shame minor offenders – for example, authorities in Fuzhou have published the names of jaywalkers caught by facial recognition and even notified their employers.[5,6]



A defining feature of China's public security surveillance is the integration of biometric identification technologies. The "Skynet" and "Sharp Eyes" systems use AI-powered facial recognition on live camera feeds to match faces against police databases. This has been expanded to essentially all spheres of public life, from scanning crowds at transit hubs to automatically verifying identities at building entrances.[7] Police experts claim such systems make finding suspects no longer *"like searching for a needle in a haystack"*, as one MPS researcher put it, because suspects can be picked out of crowds within milliseconds.[8] **Sharp Eyes**, for instance, is a rural surveillance initiative that uses AI, facial recognition and big data analytics to monitor communities, while **Safe City** is an urban



5 See https://www.theatlantic.com/international/archive/2018/02/china-surveillance/552203/, date accessed September 3, 2025
6 See https://www.nytimes.com/video/world/asia/100000008314175/china-government-surveillance-data.html, date accessed September 3, 2025
7 Ibid.
8 See https://www.cac.gov.cn/2015-06/01/c_1115476484.htm, date accessed September 3, 2025

system linking city-wide camera networks for crime and traffic control. Beijing-based firm **WATRIX** revealed another supplemental tracking system that can mobilize tens of thousands of real-time cameras to recognize a person's gait and help police nab suspects. Its pilot system is being used in some Chinese cities. **WATRIX** is a company incubated by **Institute of Automation under the Chinese Academy of Sciences (CASIA)**.[9] Together, these projects aim to **cover virtually all public spaces** and feed data into unified platforms.

Foreigners in China are also swept into these monitoring systems. Since 2017, the MPS has required that almost all foreign visitors (ages 14–70) be fingerprinted and photographed upon entry, collecting their biometric data into national database.[10] In-country, police databases automatically log foreigners' hotel check-ins, flights, train rides, and other travel records – in fact, **public security organs can obtain all hotel, airline and railway bookings**, allowing them to know where a person (including any foreign national) has traveled. Evidence shows authorities are developing tools to link this data with live surveillance feeds: for example, technologies for **automated face recognition in surveillance videos** and even **license plate identification** are used to enhance the ability to track persons of interest in real time.[11] Telecommunications providers in China must also cooperate with authorities, meaning police can access stored mobile phone location data and possibly even some real-time phone tracking (though real-time access is not fully confirmed). In short, a foreigner's movements, communications, and even hotel stays in China leave a data trail that security agencies can readily tap into.[12]

While the Ministry of State Security (MSS) does not openly discuss the specific technologies it uses for counterespionage, it can be inferred that China's vast surveillance apparatus significantly aids its spy-catching efforts. Foreign operatives in China face an environment dense with cameras, biometric scanners, and monitored digital networks. Anonymity is difficult: a foreign spy can be picked up on a facial recognition camera the moment they step outside or flagged by metadata if they contact a target. In one anecdote from January 2025, the MSS revealed how foreign agents use social media and phone apps to conduct espionage, for example by buying Chinese citizens' SIM cards and social media accounts to cover their tracks. The MSS investigated a case where a Chinese man sold his mobile phone number online, only to later discover it was purchased and used covertly by a foreign spy agency.[13] Other cases showed spies posting fake job ads to hire locals to photograph sensitive sites or attempting to ensnare Chinese students and researchers in "honey trap" relationships. The fact that state security could trace a sold phone

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9 See https://www.globaltimes.cn/content/1156769.shtml, date accessed September 3, 2025
10 See https://www.lexology.com/library/detail.aspx?g=9c7a62d9-b001-44ab-ae6c-2b85d3301b15, date accessed September 3, 2025
11 See https://www.hrw.org/zh-hans/news/2017/11/19/311556, date accessed September 3, 2025
12 Ibid.
13 See https://www.globaltimes.cn/page/202501/1326430.shtml, date accessed September 3, 2025

number or uncover online recruitment schemes implies they are leveraging electronic surveillance and data analysis to **follow the digital footprints** of foreign intelligence operations.[14]

China's Public Security and Military Apparatus Targeting IARPA Published Results

China has shown sustained interest in analyzing and reverse-engineering IARPA programs to inform their own strategic development in public security, surveillance, and military intelligence. A series of openly published publications from leading Chinese police and PLA universities reveal systematic efforts to dissect and utilize IARPA's programmatic structure, technical goals, and national security value chains.

For example, the School of Investigation and Counter-Terrorism, Chinese People's Public Security University (**中国人民公安大学**侦查与反恐怖学院), an elite training institution under the Ministry of Public Security, published multiple analyses explicitly focused on IARPA's role in counterterrorism and intelligence personnel evaluation. One paper, titled ***"Research on Performance Evaluation Method of US Intelligence Personnel: Case Study of MOSAIC Project,"***[15,16] sought to understand how the U.S. leverages behavioral analytics to assess insider threats.

A publication titled *"**Study on USA Anti-terrorism Intelligence Capability and Its Experience Based on Artificial Intelligence, Take IARPA-led research program as an Example"***[17] funded under the MPS project **"Potential Terrorist Risk Assessment and Early Warning of Key Targets"** further highlighted the MPS's intent to utilize U.S. anti-terrorism AI strategies for adaptation within China's public security apparatus, where mass-surveillance and AI surveillance tools are already pervasive. **Although such research is framed as counterterrorism, in practice, "terrorism" is often a euphemism for foreign intelligence threats, particularly American operatives and assets targeted by Chinese counterintelligence and counterespionage efforts.**

A publication authored by the researchers at the PLA Information Engineering University and the PLA's premier scientific and research institution, NUDT, have focused on IARPA's technology

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14 Ibid.
*15 Liu, J. (2019, November 15). Research on performance evaluation method of US intelligence personnel: Case study of MOSAIC Project [研究美国情报人员绩效评估方法, , 以"MOSAIC计划"为例]. People's Public Security University of China. CNKI ID: QBTS201911008. https://app.dataabyss.ai/publication/69wdQ48BmOl07zvavdw_*
16 MULTIMODAL OBJECTIVE SENSING TO ASSESS INDIVIDUALS WITH CONTEXT (MOSAIC): The MOSAIC program aimed to advance the Intelligence Community's ability to evaluate an individual's psychological drivers, cognitive abilities, and mental resilience to predict their job performance. https://www.iarpa.gov/research-programs/mosaic
17 Xuan, J., & Xiao, Y. (2019, December 5). Anti-terrorism intelligence capability and its experience based on artificial intelligence: Take IARPA-led research program as an example [基于人工智能的美国反恐情报能力及经验研究, , 以IARPA主导的研究项目为例]. Journal of Intelligence, (6), Article 003. https://doi.org/10.16231/j.cnki.jhpc.2019.06.003, https://app.dataabyss.ai/publication/1d8kQ48BmOl07zvai5Ro

forecasting and project management methodologies. A study by the PLA Information Engineering University titled ***"Research on IARPA's Project Management"***[18] examined how IARPA fosters high-risk, high-payoff innovation, with the goal of adapting similar structures to accelerate PLA technology acquisition. Similarly, a paper from NUDT and the Academy of Military Sciences evaluated IARPA's FUSE program (*Forecasting Uncertainty in Science and Engineering*)[19] to refine China's own data-driven military S&T forecasting capabilities.[20]

Finally, a publication titled ***"Research on the Organization and Management of Anticipatory Intelligence Work"*** *by* Chinese intelligence analysts studies the impact of IARPA's four flagship projects FUSE, ForeST (Forecasting Science and Technology),[21] ACE (Aggregative Contigent Estimation),[22] and OSI (Open-Source Indicators)[23] on the USIC's anticipatory intelligence work. The research details how each project was structured, the kinds of data and methods it employed, and how its outputs were evaluated. By unpacking these cases, the research identifies common processes and guiding principles, highlights the distinctive features of U.S. anticipatory-intel efforts, and offers concrete recommendations for Chinese intelligence organizations wishing to build similar capabilities.[24]

These studies reveal a deliberate effort by China's intelligence and public security apparatus, alongside military-affiliated entities, to extract lessons, methodologies, and technical knowledge from IARPA-funded programs. This includes attempts to reverse-engineer research outputs, replicate experimental designs, and adapt USIC technologies for use in China's mass surveillance apparatus and strategic military capabilities. These are not mere academic exercises. Rather, they represent a strategy to target, mirror, outpace, and ultimately neutralize U.S. intelligence and defense advantages through strategic mimicry, targeted adaptation, and systematic exploitation.

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18 Li, H., & Li, J. (2018, September 1). Research on IARPA's project management [IARPA项目管理研究]. Journal of Intelligence (情报杂志), 2018(). CNKI ID: QBZZ201809007. https://app.dataabyss.ai/publication/aKF3Qo8BmOl07zvaMk2L

19 Forecasting Uncertainty in Science and Engineering: The FUSE program seeks to develop automated methods that aid in the systematic, continuous, and comprehensive assessment of technical emergence using information found in published scientific, technical, and patent literature. FUSE is creating a system that can (1) process the massive, multi-discipline, growing, noisy, and multilingual body of scientific and patent literature from around the world; (2) automatically generate and prioritize technical terms within emerging technical areas, nominate those that exhibit technical emergence, and provide compelling evidence for the emergence; and (3) provide this capability for literature in the English and Chinese languages. https://www.iarpa.gov/research-programs/fuse

20 Luo, W., & Tian, C. (2019, January 31). Review and analysis of data-driven technology forecasting: A case study of the FUSE project [数据驱动的技术预测之研究评析, 以FUSE项目为例]. Information Studies: Theory & Application (情报理论与实践), 42(7). https://doi.org/10.16353/j.cnki.1000-7490.2019.07.003, https://app.dataabyss.ai/publication/rKiOQo8BmOl07zvaDa4d

21 See https://www.iarpa.gov/research-programs/forest, date accessed September 3, 2025

22 See https://www.iarpa.gov/research-programs/ace, date accessed September 3, 2025

23 See https://www.iarpa.gov/research-programs/osi, date accessed September 3, 2025

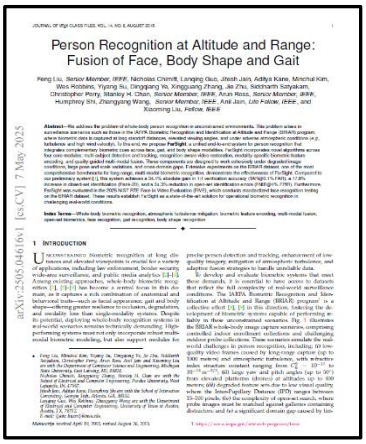
24 Li Guojun　Wang Yanfei　Xu Yang. Research on the Organization and Management of Anticipatory Intelligence Work (美国预见性情报工作的组织管理及启示研究). Journal of Information Resources Management, 2024, 14(3): 80-89, https://jirm.whu.edu.cn/jwk3/xxzyglxb/EN/10.13365/j.jirm.2024.03.080

PRC Access to IARPA Research: Case Studies in High-Risk Collaborations

For nearly a decade, IARPA-funded research has been conducted alongside some of the highest-risk institutions within China's defense research and industrial base. These collaborations span multiple IARPA programs and technology areas central to intelligence and military advantage, including speech recognition, synthetic biology, cyber operations, quantum science, artificial intelligence, and advanced image reconstruction. The scope of these activities shows a failure to prevent U.S. government–sponsored research from intersecting with PRC entities that pose clear national security risks. Moreover, many of the publications documented in the following section are recent, and several were published after the Chinese institutions involved had already been added to various U.S. government entity lists for national security reasons. That pattern indicates that existing safeguards, policies, and vetting mechanisms have been insufficient to prevent U.S. taxpayer-funded research from flowing directly China's defense research and industrial base.

Case Study 1: *Deep-Dive into Michigan State University's IARPA BRIAR Program*[25]

The IARPA BRIAR program began in November 2021, with the goal of developing software algorithm-based systems capable of performing whole-body biometric identification at long-range and from elevated platforms. The program states many USIC agencies require the ability to identify or recognize individuals under challenging scenarios, such as at long-range, through atmospheric turbulence, or from elevated and/or aerial sensor platforms (e.g., ≥20° sensor view angle from watchtowers or unmanned aerial vehicles).[26]


JOURNAL OF IEEE CLASS FILES, VOL. 14, NO. 8, AUGUST 2015

Person Recognition at Altitude and Range: Fusion of Face, Body Shape and Gait

Feng Liu, Senior Member, IEEE, Nicholas Chimitt, Lanqing Guo, Jitesh Jain, Aditya Kane, Minchul Kim, Wes Robbins, Yiqing Su, Dingqiong Ye, Xingguang Zhang, Jie Zhu, Siddharth Satyakam, Christopher Perry, Stanley H. Chan, Senior Member, IEEE, Arun Ross, Senior Member, IEEE, Humphrey Shi, Zhangyang Wang, Senior Member, IEEE, Anil Jain, Life Fellow, IEEE, and Xiaoming Liu, Fellow, IEEE

arXiv:2505.04616v1 [cs.CV] 7 May 2025

1 INTRODUCTION

To develop and evaluate biometric systems that meet these demands, it is essential to have access to datasets that reflect the full complexity of real-world surveillance conditions. The IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program[1] is a large-scale effort to fill this critical gap and support the development of biometric systems capable of performing reliably in these unconstrained scenarios. Fig. 1 illustrates the BRIAR multi-body image capture scenarios, comprising controlled indoor enrollment collections and challenging outdoor probe collections. These scenarios simulate the real-world challenges in person recognition, including: (i) low-quality video frames caused by long-range capture (up to 500 meters) and atmospheric turbulence, with refractive index structure constant ranging from $C_n^2 = 10^{-17}$ to 10^{-14} $m^{-2/3}$; (ii) large yaw and pitch angles (up to 90°) from elevated platforms (drones at altitudes up to 400 meters); (iii) degraded feature sets due to low visual quality and small face sizes (typically fewer than 20 pixels, 25-200 pixels for the majority of cases in capture, where probe images must be matched against galleries containing distractors; and for a significant domain gap caused by low-

[1] https://www.iarpa.gov/research-programs/briar

As recently as May 2025, IARPA funded research titled **"Person Recognition at Altitude and Range: Fusion of Face, Body Shape and Gait"** focuses on long-range person recognition in challenging surveillance environments, such as high altitudes and long distances with poor atmospheric conditions. The system, called **FarSight**, fuses facial features, body shape, and gait using AI to identify individuals even when image quality is degraded. It's part of the **IARPA BRIAR program**, aimed at enhancing biometric recognition capabilities for intelligence and defense operations under real-world, high-risk conditions.[27]

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25 BIOMETRIC RECOGNITION AND IDENTIFICATION AT ALTITUDE AND RANGE (BRIAR): The BRIAR program aims to provide the U.S. Intelligence Community with the ability to perform accurate and reliable biometric identity intelligence across a wider range of imagery and collected from a wider selection of sensor platforms. 25 See https://www.iarpa.gov/research-programs/briar, date accessed October 7, 2025
26 Ibid.
27 See https://arxiv.org/abs/2505.04616, date accessed September 3, 2025

The $11M IARPA BRIAR program is led by a professor from the Department of Computer Science and Engineering at **Michigan State University** (MSU) and includes collaborators from the University Texas at Austin, Purdue University and the University of Oregon.[28] The MSU professor has a long history of supporting IARPA programs while simultaneously collaborating with institutions like **Southern University of Science and Technology (SUSTech)**, in China, on similar gait research.[29] [30] [31] [32] [33] The MSU professor's 2024 research with SUSTech, funded under National Natural Science Foundation of China (NSFC) award 61976144, involved **Galaxy Water Drop Technology (Beijing) Co., Ltd. (银河水滴科技(北京)有限公司), aka WATRIX**, as co-investigators on the project "Research on large-scale gait recognition based on incomplete gait data". WATRIX develops gait big data analysis platforms for public security surveillance in China and is known to have the world's highest gait recognition accuracy and the largest gait database.[34] [35] WATRIX was incubated by the **Institute of Automation of the Chinese Academy of Sciences (CASIA),** who the professor also collaborates with[36], and co-founded by Academician Tan Tieniu (谭铁牛).[37] According to The Hoover Institution's report Eye's Wide Open, Tan Tieniu is a "senior CASIA leader and renowned expert on computer vision and biometric recognition, collaborates on research internationally while overseeing domestic public security and surveillance projects. Tan concurrently serves as a deputy director of the PRC's government liaison office in Hong Kong. The US Treasury Department recently added Tan to its "specially designated nationals list" as part of the US government's sanctions on Hong Kong officials for their responsibility for a human rights crackdown in the city." The MSU professor's gait research with SUSTech, CASIA, and other PRC institutions illustrates how USIC funding can end up directly benefiting China's surveillance state and human rights abuses.[38]

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28 See https://engineering.msu.edu/news-events/news/2023/01/25/ieee-names-xiaoming-liu-prestigious-fellows-list-2023, date accessed September 3, 2025
29 See https://www.mlive.com/news/2023/10/msu-researchers-are-working-with-us-intelligence-agencies-on-identification-technology.html, date accessed September 3, 2025
30 See https://statenews.com/article/2023/10/msu-researchers-developing-long-range-identification-technology-for-us-intelligence-community?ct=content_open&cv=cbox_latest, date accessed September 3, 2025
31 See https://www.iarpa.gov/images/research-programs/Video%20LINCS/Proposers%20Day/LT07%20Michigan%20State%20-%20Liu.pdf, date accessed September 3, 2025
32 See https://arxiv.org/pdf/2402.19122, date accessed September 3, 2025
33 See https://arxiv.org/pdf/2505.18132, date accessed September 3, 2025
34 See http://www.watrix.ai/product/foresight/?productType=0, date accessed September 3, 2025
35 See https://baike.baidu.com/item/%E9%93%B6%E6%B2%B3%E6%B0%B4%E6%BB%B4%E7%A7%91%E6%8A%80%EF%BC%88%E6%B1%9F%E8%8B%8F%EF%BC%89%E6%9C%89%E9%99%90%E5%85%AC%E5%8F%B8/64280300, date accessed September 3, 2025
36 See https://arxiv.org/pdf/2505.13327, date accessed September 3, 2025
37 Ibid.
38 See https://www.hoover.org/sites/default/files/research/docs/stoff-tiffert_eyeswideopen_web_revised.pdf, date accessed September 3, 2025

Additionally, the MSU professor has extensive connections in China's Computer Vision ecosystem and actively collaborates with Shanghai Jiao Tong University and Shanghai Artificial Intelligence Laboratory.[39] [40] The professor's academic visits and exchanges in China were numerous, many of which overlapped with the periods in which he was receiving support from IARPA. Example summary of the MSU professor's visits and academic exchanges in China include:

- **April 2025 – Nanjing University**: Gave a lecture on *Visual Recognition in the Era of Foundation Models* as part of the "Frontier Series on Intelligent Science and Technology."[41]

- **June 2025 – Tongji University:** Gave a lecture titled "Computer Vision: A Journey of Pursuing 3D World Understanding" to Tongji students. The lecture aimed to explore the impact of cutting-edge 3D visual achievements, from 3D modeling to 3D reconstruction, on the field of computer vision.[42]

- **November 2023 – ShanghaiTech University**: Delivered a SIST Distinguished Seminar on *Computer Vision and 3D World Understanding*.[43]

- **December 2023 – Chinese Conference on Biometric Recognition (CCBR)**: Participated in China's premier biometrics-focused conference where he was a special guest speaker. The conference is sponsored and organized by Chinese companies and institutions such as the Institute of Automation, Chinese Academy of Sciences (CASIA) a highly concerning partner of public security organs with ties to mass surveillance technologies associated with human rights violations, particularly in the Xinjiang region. [44] [45]

- **December 2016 – Beihang University (BUAA)**: Visited the Intelligent Recognition and Image Processing Lab for an academic lecture he gave titled "2D/3D Shape Estimation and Recognition for Large-pose Faces." After the presentation, he toured the Intelligent Recognition and Image Processing Laboratory.[46] Beihang University is a State Administration of Science, Technology and Industry for National Defense (SASTIND)-administered university.[47]

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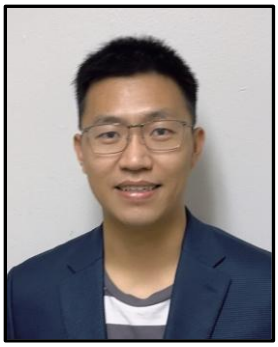
39 See https://cvlab.cse.msu.edu/pdfs/song_dvf_nip2024.pdf, date accessed September 3, 2025
40 See https://arxiv.org/pdf/2503.20188, date accessed September 3, 2025
41 See https://is.nju.edu.cn/is_en/92/c5/c57786a758469/page.htm, date accessed September 3, 2025
42 See https://vill.tongji.edu.cn/info/1056/2066.htm, date accessed September 3, 2025
43 See https://shanghaitech.edu.cn/eng/2023/1128/c2445a1086342/page.htm, date accessed September 3, 2025
44 See https://ccbr99.cn/2023/, date accessed September 3, 2025
45 See https://www.hoover.org/sites/default/files/research/docs/stoff-tiffert_eyeswideopen_web_revised.pdf, date accessed September 3, 2025
46 See https://irip.buaa.edu.cn/info/1066/1598.htm, date accessed September 3, 2025
47 See https://www.aspi.org.au/report/china-defence-universities-tracker/, date accessed November 24, 2025

- **December 2016 – Horizon Robotics:** Delivered a technical lecture at Horizon Robotics' "Big Bull Lecture Hall," where he presented his research on face recognition, large-pose 2D/3D shape estimation, and multi-angle biometric identification using deep learning and 3D face models. The presentation focused on methods with direct applicability to facial recognition, surveillance, and identity tracking in unconstrained, real-world environments.[48] Horizon Robotics was referenced by the House Select Committee on the CCP in a letter to GSR Ventures, noting that Horizon's CEO was publicly recognized in October 2017 for contributions to the PRC's military-civil fusion strategy.[49]

- **July 2015 – Zhejiang University**: Presented an academic lecture on *Unconstrained 3D Face Reconstruction*.[50]

- **October 2013 – Zhiyuan Community (online)**: Participated in a closed-door AI session on anti-fraud and intelligent risk control.[51]



Feng Liu, postdoctoral researcher trained under the MSU professor, conducted research under a BRIAR project titled "Physics-driven Modeling and Learning for Person Recognition at a Distance and Altitude." Noteworthy, in 2017-2018 he was funded in China for a project called **"Research on Key Technologies of 3D Face Reconstruction in Public Security Surveillance and Recognition"** in which he was collaborating with Michigan State (2019) and was likely his entry connection into the U.S.[52,53] This PRC public security project was funded by **The Miaozi Key Project in Science and Technology Innovation Program of Sichuan Province**, a project closely tied to Sichuan's Public Security apparatus focused on 3D face reconstruction from mugshots.[54] Liu's trajectory from Sichuan University's public security–linked facial reconstruction program to IARPA-sponsored long-range biometric recognition represents a clear technology transfer risk, with skills and methods readily adaptable to PRC surveillance and counterintelligence operations.

The MSU professor's sustained collaboration with PRC universities and surveillance-linked research bodies underscores a high-probability vector for sensitive U.S.-funded intelligence technologies to transition into China's state security and defense programs, where they can be operationalized for both domestic control and foreign intelligence targeting.

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48 *See https://archive.ph/hdcoU, date accessed February 4, 2026*
49 *See https://chinaselectcommittee.house.gov/sites/evo-subsites/selectcommitteeontheccp.house.gov/files/evo-media-document/2023.07.18-letter-to-gsr.pdf, date accessed February 5, 2026*
50 *See http://www.cs.zju.edu.cn/csen/2015/0730/c27029a1296293/page.htm, date accessed September 3, 2025*
51 *See https://hub.baai.ac.cn/view/31516, date accessed September 3, 2025*
52 *See https://www.sciencedirect.com/science/article/abs/pii/S0925231220309279, date accessed September 3, 2025*
53 *See https://drexel.edu/~/media/Files/cci/docs/Faculty/CV_FengLiu.ashx, date accessed September 3, 2025*
54 *See https://www.sciencedirect.com/science/article/abs/pii/S0925231220309279, date accessed September 3, 2025*

The risk is not abstract: Chinese institutions exist within a national innovation system explicitly built to serve MCF strategy. When IARPA-funded researchers originating from these institutions return, or maintain ties through joint patents, remote platforms, or ongoing academic collaborations, they do so with elevated value to the MSS, MPS, or PLA. Their U.S.-sponsored training becomes a strategic asset to China's internal security agencies and defense establishment. Additionally, exposure to sensitive biometric architectures and U.S. intelligence R&D methodologies may enable more sophisticated counterintelligence, surveillance, and censorship systems, tailored to detect, track, and neutralize U.S. assets or suppress domestic dissent.

Given the systematic overlap between the technical goals of IARPA and the mission mandates of these Chinese security-aligned research entities, the reintegration of these personnel represents a **clear vector for adversarial innovation, strategic exploitation, and long-term risk to U.S. national security**.

Case Study 2: November 2025 – *Hidden Backdoor Attack against Neural Code Search Models*
This IARPA-funded TrojAI project, under contract W911NF-19-S-0012, published publication on hidden backdoor attack against neural code search models research was co-authored with researchers from Purdue University, Nanjing University's State Key Laboratory of Novel Software, and Nanjing University of Science and Technology, one of China's "Seven Sons of National Defense" schools.[55]

- State Key Laboratory of Novel Software (计算机软件新技术国家重点实验室) conducts research on software methodology, machine learning and intelligent information processing technology, software quality assurance and automation technology, distributed computing and system security technology, media computing and content processing technology. The Lab has received funding from China's 973 and 863 Programs.[56] Additionally, the lab's academic committee includes representatives from China Aerospace Science and Technology Corporation, the PLA's NUDT, Beihang University, and the Academy of Military Sciences. Every one of these entitles sits at the core of China's defense research ecosystem, underscoring the lab's direct and sustained ties to PRC military, state-owned enterprise defense conglomerates, and security entities.[57]

55 See Journal Publication, Yuchen Chen et al, https://doi.org/10.1145/3774421, accessed on November 23, 2025
56 See Website, Nanjing University, State Key Laboratory of Novel Software and Technology, Brief introduction of the Laboratory, https://keysoftlab.nju.edu.cn/1583/list.htm, accessed on November 23, 2025
57 See Website, Nanjing University, State Key Laboratory of Novel Software and Technology, Academic Committee, https://keysoftlab.nju.edu.cn/1585/list.htm, accessed on November 23, 2025

Case Study 3: December 2024 – *Leveraging Thermal Modality to Enhance Reconstruction in Low-Light Conditions*

This IARPA-funded project, under contract 140D0423C0076, explores advanced techniques to enhance image reconstruction performance in low-light environments by incorporating thermal modalities. This technology has direct applications in surveillance, reconnaissance, and nighttime operations, domains of high strategic importance to the USIC. The research was conducted jointly by Johns Hopkins University and the **Southern University of Science and Technology (SUSTech) in Shenzhen, China**, with researcher Mingqian Liao representing SUSTech. SUSTech is a rapidly rising Chinese university with documented partnerships across China's defense research ecosystem.[58] The inclusion of a PRC-based researcher in a project that advances low-light and thermal image reconstruction raises significant concerns due to its dual-use potential in covert surveillance, autonomous targeting, and low-visibility ISR systems.[59]

Case Study 4: September 2024 – *Ultrafast Hybrid Computing Systems Enabled by Memristor-Based Quadratic Programming Circuits*

This research, funded by IARPA under contract 2021–210900005 and partially supported by the Air Force Research Laboratory (AFRL) through its Conduction Channel Engineering in Memristor for Synapse program (Grant FA-8750-19-1-0503), presents a high-performance computing architecture based on memristor-enabled quadratic programming circuits. The project, led primarily by the University of Southern California with co-authors from the University of Waterloo and the University of Virginia, also includes Boxiang Song of **Wuhan National Laboratory for Optoelectronics (**WNLO, 武汉光电国家实验室**) at Huazhong University of Science and Technology (HUST)**, a Chinese institution with deep ties to the PRC's military-civil fusion and optoelectronics weapons research programs and co-administered by SASTIND. HUST houses national defense laboratories and is a well-known collaborator with defense-linked state-owned enterprises.

- WNLO was one of six national research centers approved by China's Ministry of Science and Technology in 2017. WNLO conducts advanced research in optoelectronics with applications spanning information science, energy technologies, and life sciences. Its core areas include ultrafast lasers, laser-based manufacturing, optoelectronic device integration, and data storage technologies, all of which have substantial dual-use and military relevance. WNLO is jointly organized by three primary entities: the Wuhan Institute of Physics and Mathematics (WIPM) under the Chinese Academy of Sciences; the Wuhan Research Institute of Post and Telecommunications (WRIPT); and most

58 See https://unitracker.aspi.org.au/universities/southern-university-of-science-and-technology/
59 Xu, J., Liao, M., Kathirvel, R.P., Patel, V.M. (2025). Leveraging Thermal Modality to Enhance Reconstruction in Low-Light Conditions. In: Leonardis, A., Ricci, E., Roth, S., Russakovsky, O., Sattler, T., Varol, G. (eds) Computer Vision – ECCV 2024. ECCV 2024. Lecture Notes in Computer Science, vol 15095. Springer, Cham. https://doi.org/10.1007/978-3-031-72913-3_18

notably, the 77th Research Institute of the China Shipbuilding Industry Corporation (CSIC), a military research entity under China's state-owned defense conglomerate.[60]

- In 2006, WNLO was approved by the State Administration of Foreign Experts Affairs (now part of the Ministry of Science and Technology) and the Ministry of Education under China's "111 Program.[61] Also known as the "Program for Introducing Talents to Universities", the 111 Program is a Chinese state-sponsored talent recruitment initiative designed to bring in high-level foreign experts to support strategic disciplines, often with dual-use fields.[62] WNLO's participation in this program reflects its alignment with China's national strategy to absorb foreign scientific talent, human capital, and technology in areas like advanced optoelectronics, which are directly relevant to both commercial and defense applications.[63]
- In 2008, WNLO was approved by the Central Organization Department of the Chinese Communist Party as an "Innovation and Entrepreneurship Base for Overseas High-Level Talents" (海外高层次人才创新创业基地).[64] Innovation and Entrepreneurship Bases are PRC-designated platforms designed to attract and integrate overseas high-level talent, particularly those with backgrounds in cutting-edge science and technology disciplines. These bases are typically tasked with introducing original, innovative, and strategically significant foreign high-tech projects into China. Their core functions include showcasing the latest overseas S&T achievements and products, facilitating technology transfer, and serving as a bridge for foreign-trained experts to commercialize or operationalize their research in China.[65]
- In 2011, WNLO established the Photon Detection and Radiation Functional Laboratory in the 717th Research Institute of CSIC.[66]
- In 2013, HUST/WNLO signed a cooperative framework agreement with the China Academy of Launch Vehicles Technology, a core entity within China's state-owned space and missile industry subordinate to state-owned defense conglomerate CASC, and Capital Aerospace Machinery Company. As part of this agreement, the parties jointly established a Joint Laboratory of Additive Manufacturing Technology.[67]

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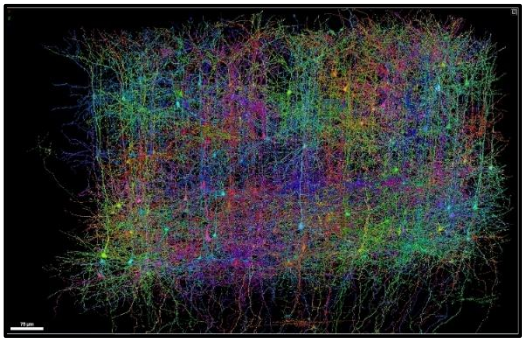
60 See https://selectcommitteeontheccp.house.gov/sites/evo-subsites/selectcommitteeontheccp.house.gov/files/evo-media-document/fox-in-the-henhouse-report-compressed_0.pdf, date accessed November 23, 2025
61 See Website, Wuhan National Laboratory for Optoelectronics, Historical Evolution, https://wnlo.hust.edu.cn/gywm/lsyg.htm
62 Ibid.
63 Ibid.
64 Ibid.
65 See Public Release, Chinese Overseas Talents Innovation and Entrepreneurship Competition, 2024, https://cotiec.cast.org.cn/index.php?m=content&c=index&a=lists&catid=50
66 Ibid.
67 Ibid.

This research, which leverages nanoscale fabrication, analog computing, and neuromorphic hardware, has significant dual-use implications for advanced signal processing, cryptographic acceleration, and battlefield AI decision-making. The integration of a PRC researcher into an ODNI and AFRL-backed program in next-generation hybrid computing poses a clear strategic risk, potentially exposing sensitive R&D at the forefront of U.S. defense computing innovation to adversarial appropriation.[68]

Case Study 5: July 2024 – *Coherent Coupling and Non-Destructive Measurement of Trapped-Ion Mechanical Oscillators*

This research, funded by IARPA and the National Institute of Standards and Technology (NIST) Quantum Information Program, explores precision measurement and quantum control techniques in trapped-ion systems, an essential foundation for quantum computing, secure communications, and advanced sensing. Conducted primarily at the NIST and the University of Colorado Boulder, the project also involves Pan-Yu Hou, whose current institutional affiliation is with the **Center for Quantum Information at Tsinghua University (a SASTIND co-administered university)**. While Pan-Yu Hou's contribution occurred during their time at NIST, their subsequent transition to Tsinghua University's quantum program raises serious concerns about knowledge transfer and exposure of sensitive quantum research funded by IARPA to a strategic adversary. The work's implications in coherent quantum control and non-destructive measurement directly support applications in quantum surveillance, encrypted communications, and quantum-enhanced sensors, areas in which China is aggressively competing for technological dominance.[69]



IARPA Intelligence Value "The MICrONS program aimed to close the performance gap between human analysts and automated pattern recognition systems by reverse-engineering the algorithms of the brain"

Case Study 6: August 2024 – *Multiplexed In Situ Protein Imaging Using DNA-Barcoded Antibodies with Extended Hybridizatineurological targeting* research project, funded in part by IARPA under the MICrONS (Machine Intelligence from Cortical Networks) Program[70] (D16PC0008), advances high-resolution protein imaging technologies by combining DNA-barcoded antibodies with enhanced hybridization chain reactions, an innovation with major implications for neuroscience, cellular mapping, and potentially neurological targeting or brain-computer

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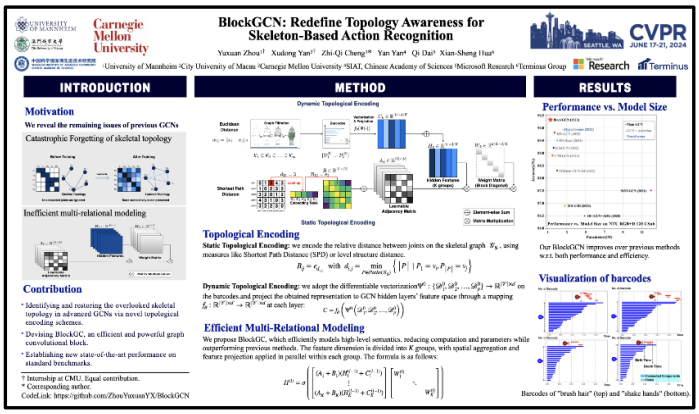
68 Z. Liu, H.-C. Cheng, S. Hossain, D. Meng, R. Bena, Y. Shi, B. Chen, D. W. Yang, S. Su, Y. Wang, P. Hu, M. Palaria, H. Yang, Q. Zhang, B. Song, T.-H. Ou, J. Ye, N. T. Hiramony, H. Zhang, T.-H. Hsu, Z. Tang, Z. Cai, M. Barnell, Q. Wu, C. Yang, S. Cronin, Q. Nguyen, M. S.-W. Chen, W. Wu, Ultrafast Hybrid Computing Systems Enabled by Memristor-Based Quadratic Programming Circuits. Adv. Funct. Mater. 2024, 34, 2401600. https://doi.org/10.1002/adfm.202401600
69 Hou, PY., Wu, J.J., Erickson, S.D. et al. Coherent coupling and non-destructive measurement of trapped-ion mechanical oscillators. Nat. Phys. 20, 1636–1641 (2024). https://doi.org/10.1038/s41567-024-02585-y
70 See https://www.iarpa.gov/research-programs/microns, date accessed September 3, 2025

interface development. The work was led by researchers at the Wyss Institute and Harvard Medical School. However, the collaboration includes Yu Wang and Xiaoyu Liu, both affiliated with the **Key Laboratory of Quantitative Synthetic Biology at the Shenzhen Institute of Advanced Technology (SIAT), Chinese Academy of Sciences (CAS)**, a PRC national laboratory. The project also received funding from the Chinese Academy of Sciences, Fudan University, and Tsinghua University. Given the potential applications in biological sensing, cognitive mapping, and synthetic neurobiology, the collaboration poses significant national security concerns.[71]

Case Study 7: June 2024 – *BlockGCN: Redefine Topology Awareness for Skeleton-Based Action Recognition*

This project, funded in part by IARPA under contract D17PC00340, as well as by Defense Advanced Research Projects Agency (DARPA), AFRL, and the U.S. Department of Commerce, develops advanced Graph Convolutional Network (GCN) models for skeleton-based human action recognition, a field with direct applications in surveillance, autonomous systems, and behavioral prediction. The research team spans Carnegie Mellon University, Microsoft Research, the University of Mannheim, and the City University of Macau and the **Shenzhen Institute of Advanced Technology (SIAT), Chinese Academy of Sciences**, with Yan Yan representing the latter. SIAT, a key institute under the Chinese Academy of Sciences, plays a central role in China's national AI strategy and maintains ties to defense-focused research, particularly in computer vision and human-machine interaction.[72] The technology developed in this project has dual-use potential, including use in automated surveillance, crowd behavior analysis, and motion recognition for both civil and military applications. The inclusion of SIAT, a known entity in China's military-civil fusion ecosystem, on a U.S. intelligence and defense-funded AI project raises serious security and export control concerns, particularly given China's ambitions to dominate next-generation surveillance.[73,74]



Poster Published in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

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71 Yu Wang, Xiaoyu Liu, Yitian Zeng, Sinem K Saka, Wenxin Xie, Isabel Goldaracena, Richie E Kohman, Peng Yin, George M Church, Multiplexed in situ protein imaging using DNA-barcoded antibodies with extended hybridization chain reactions, Nucleic Acids Research, Volume 52, Issue 15, 27 August 2024, Page 71, https://doi.org/10.1093/nar/gkae592
72 See https://www.siat.ac.cn/jggl/jgjj/, date accessed November 24, 2025
73 Y. Zhou, X. Yan, Z. -Q. Cheng, Y. Yan, Q. Dai and X. -S. Hua, "BlockGCN: Redefine Topology Awareness for Skeleton-Based Action Recognition," 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024, pp. 2049-2058, doi: 10.1109/CVPR52733.2024.00200.
74 See https://cvpr.thecvf.com/virtual/2024/poster/30473, date accessed September 3, 2025

Case Study 8: June 2024 – *UniGS: Unified Representation for Image Generation and Segmentation*

This IARPA-funded project, supported under contract 140D0423C0074, develops a unified framework for image generation and segmentation, enabling machines to both synthesize visual content and interpret it with high spatial precision, capabilities crucial to surveillance, scene understanding, and cognitive AI systems. Led by researchers at the University of California, Merced and The University of Sydney, the project also involves Weidong Guo and Yu Xu from the **QQ Browser Lab at Tencent**, and Bo Du from **Wuhan University (a SASTIND co-administered school)**. Tencent, one of China's most powerful tech conglomerates, maintains direct partnerships with state-aligned entities and is subject to China's national security and data sharing laws. Wuhan University is also closely tied to China's military-civil fusion framework and has hosted national defense laboratories involved in remote sensing and AI.[75] The participation of Tencent-affiliated researchers and a PLA-linked university in this IARPA-supported effort underscores a critical failure to screen foreign commercial and academic collaborators for national security risk. Image generation and segmentation systems have high-risk dual-use applications, including synthetic media creation, satellite image interpretation, and automated surveillance feeds.[76]

Case Study 9: September 2024 – *Towards General Robustness Verification of MaxPool-Based Convolutional Neural Networks via Tightening Linear Approximation*

This research, partially funded by IARPA through Army Research Office contract W911NF-19-S-0012, advances methods for verifying the robustness of convolutional neural networks (CNNs), specifically those using MaxPooling operations, by proposing tighter linear approximations. Robustness verification is essential for secure AI systems in critical applications such as autonomous systems, defense analytics, and adversarial attack resistance. While the project involves U.S.-based researchers from the University of Massachusetts Amherst and Penn State, it is co-led by Yuan Xiao, Chunrong Fang, and Zhenyu Chen from the **State Key Laboratory for Novel Software Technology at Nanjing University**. This work was also funded by the National Natural Science Foundation of China and the Shenzhen Municipal Science and Innovation Commission. Given the importance of CNN robustness to national defense, especially in contested electronic or AI-driven environments, this project raises significant research security concerns.[77]

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75 See https://unitracker.aspi.org.au/universities/wuhan-university/, date accessed September 3, 2025
76 L. Qi et al., "UniGS: Unified Representation for Image Generation and Segmentation," 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024, pp. 6305-6315, doi: 10.1109/CVPR52733.2024.00603.
77 Y. Xiao, S. Ma, J. Zhai, C. Fang, J. Jia and Z. Chen, "Towards General Robustness Verification of MaxPool-Based Convolutional Neural Networks via Tightening Linear Approximation," 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024, pp. 24766-24775, doi: 10.1109/CVPR52733.2024.02339.

Case Study 10: January 2024 – *3D-Aware Neural Body Fitting for Occlusion Robust 3D Human Pose Estimation*

This project, supported by IARPA contract 2022-21102100005, along with funding from the Army Research Laboratory, National Institutes of Health (NIH), and the Office of Naval Research, focuses on occlusion-robust 3D human pose estimation, a critical technology for surveillance, human behavior analysis, and autonomous system tracking. The research is primarily led by Johns Hopkins University, but it also includes Pengliang Ji from **Beihang University**, one of China's top defense-oriented institutions and a core member of the "Seven Sons of National Defense" (国防七子) universities that directly support the People's Liberation Army. Beihang University has been on the U.S. Bureau of Industry and Security Entity List since 2001 due to its involvement in rocket systems and unmanned air vehicle activities.[78] The integration of a Beihang-affiliated researcher into a project backed by IARPA and multiple U.S. defense research agencies represents a serious research security concern. Technologies capable of 3D pose estimation under occlusion are of high strategic value for both surveillance and adversarial combat applications, including drone-based ISR, smart targeting, and gait-based identification.[79]

Case Study 11: **December 2023 – *Quantum Error Mitigation***

This project funded by IARPA's LogiQ program[80], includes a researcher from the **China Academy of Engineering Physics (CAEP)**, China's primary nuclear weapons research and development complex has been on the U.S. BIS Entity List since 1997.[81] CAEP is subordinate to the Central Military Commission, the CCP's top military organ.[82] This institution's involvement in IARPA's LogiQ program[83] means sensitive U.S. quantum error correction innovations may have been exposed to the PRC's nuclear and quantum military ecosystem.

IARPA's LogiQ Enables fault-tolerant quantum computing for intelligence problem-solving.

Case Study 12: **August 2023 – *Quantum Simulation of Conical Intersections Using Trapped Ions***

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78 See Federal Register, Department of Commerce Bureau of Industry and Security, Entity List: Revisions and Additions, May 14, 2001, https://www.federalregister.gov/documents/2001/05/14/01-12188/entity-list-revisions-and-additions
79 Y. Zhang, P. Ji, A. Wang, J. Mei, A. Kortylewski and A. Yuille, "3D-Aware Neural Body Fitting for Occlusion Robust 3D Human Pose Estimation," 2023 IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2023, pp. 9365-9376, doi: 10.1109/ICCV51070.2023.00862.
80 Cian, Z., Endo, S., Brown, S. C., et al. (2023). Quantum error mitigation. Reviews of Modern Physics, 95(4), 045005. https://doi.org/10.1103/revmodphys.95.045005
81 See Public Announcement, Federal Register/National Archives, Revisions to the Export Administration Regulations: Additions to the Entity List, June 30, 1997, https://www.govinfo.gov/content/pkg/FR-1997-06-30/pdf/97-17146.pdf
82 See News Article, Derdre Kirsten Tatlow, Newsweek, How U.S. Scientists are Collaborating with China's Military: 'Wake-Up Call', January 12, 2023, https://www.newsweek.com/how-us-scientists-are-collaborating-chinas-military-wake-call-1773079
83 See https://www.iarpa.gov/research-programs/logiq, date accessed September 3, 2025

This project funded by IARPA via Army Research Office contract W911NF-16-1-0082, includes a researcher from **University of Science and Technology of China (USTC)**, a leading PLA-affiliated quantum science hub. The study shows that trapped ions can mimic molecular behavior with enough control to directly observe the geometric phase at conical intersections, something that real molecules can't reveal because the processes happen too fast to measure.[84] Additionally, although this publication predates the May 2024 addition of the University of Science and Technology of China (USTC) to the Bureau of Industry and Security (BIS) Entity List, it remains highly relevant for any future research relationships with USTC. USTC was added to the Entity List for acquiring and attempting to acquire U.S.-origin items in support of China's quantum technology capabilities, which carries serious national security implications given the military applications of quantum technologies and their role in advancing China's nuclear weapons program.[85]

Case Study 13: August 2022 – Roadmap on Soft Robotics

This project was funded by IARPA Grant 2019-19020100001, and DARPA Grant FA8750-20-C-0075 features multiple researchers from Beihang University. The roadmap highlights how advances in multifunctional materials such as programmable elastomers, stimuli-responsive composites, and bioinspired structural polymers, are enabling new generations of soft robots capable of morphing, self-adaptation, and resilient locomotion across complex environments.[86]

Case Study 14: **July 2018 – Advanced recurrent network-based hybrid acoustic models for low resource speech recognition**

A 2018 publication on hybrid-acoustic models for low research speech recognition was co-authored by researchers from the University of Kentucky, Tsinghua University, and Wei-Wei Liu of PLA Unit 62315.[87] The authors acknowledged that IARPA BABEL[88] program datasets were used to conduct the research and experiments. Wei-Wei Liu of PLA Unit 62315 concurrently filed a related patent titled "The modeling method and system of a kind of English acoustic model applied to languages identification" with other members of the same unit.[89] Although extensive searches of Chinese-language sources did not yield public information on the internal mission or structure of PLA Unit 62315, its role using IARPA-provided datasets,

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84 Whitlow, J., Jia, Z., Wang, Y. et al. Quantum simulation of conical intersections using trapped ions. Nat. Chem. 15, 1509–1514 (2023). https://doi.org/10.1038/s41557-023-01303-0

85 Public Notice, Federal Register/National Archives, Additions of Entities to the Entity List, May 14, 2024, https://www.federalregister.gov/documents/2024/05/14/2024-10485/additions-of-entities-to-the-entity-list

86 Dario, P., Rus, D., Kim, S., et al. (2022). Roadmap on soft robotics: multifunctionality, adaptability and growth without borders. Multifunctional Materials, 5(1), 012001. https://doi.org/10.1088/2399-7532/ac4c95

87 See Journal Publication, Jian Kang et al, Advanced Recurrent Network-Based Hybrid Acoustic Models for Low Resource Speech Recognition, https://asmp-eurasipjournals.springeropen.com/articles/10.1186/s13636-018-0128-6#Ack1, accessed November 23, 2025

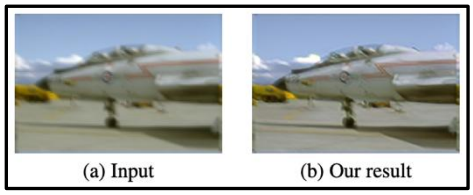
88 IARPA BABEL program develops agile, robust speech-recognition technology that can be quickly adapted to any human language, enabling analysts to rapidly search and process massive volumes of real-world recorded speech.

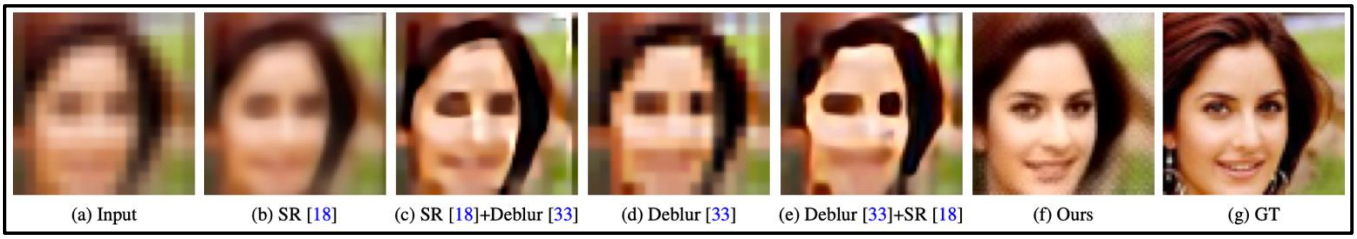
89 *See https://patents.google.com/patent/CN108510978A/, accessed February 3, 2026*

combined with the concurrent filing of a related patent by the same unit, demonstrates a clear PLA interest in advancing speech recognition and acoustic modeling capabilities. Taken together, the publication, dataset usage, and patent activity strongly suggest that this research was intended to support military or intelligence applications.

Case Study 15: December 2017 – *Learning to Super-Resolve Blurry Face and Text Images*

A 2017 publication on methods to resolve blurry face and text image research was coauthored by researchers from Harvard University, the University of California-Merced, NVIDIA, Tsinghua University, and Nanjing University of Science and Technology.[90] The research was funded by IARPA contract D16PC00002, NSF, and the China Scholarship Council (CSC). The research demonstrates the ability to reconstruct heavily distorted images into high-resolution outputs. This capability has clear applications for surveillance, identification, and other state security operations.





- The CSC functions as a state-directed talent pipeline, selecting the top echelon of Chinese students not simply for academic performance but for their alignment with national strategic priorities. Its programs are tightly integrated into China's technology acquisition system and concentrate on STEM disciplines that mirror the CCP's statutory plans, Five-Year Plans, and broader S&T directives. Most CSC recipients pursue graduate or postdoctoral training, placing them inside U.S. labs and research programs, often those supported by federal agencies. Investigative reviews show that CSC-sponsored students have gained access to sensitive U.S. government–funded research and later moved into roles within China's defense research and industrial base. CSC awardees are also obligated to maintain contact with PRC embassies and consulates, providing updates on their research focus, advisors, and institutional affiliations. Document production and publication analysis confirm repeated instances in which CSC-funded individuals conducted federally funded research while maintaining formal ties to PRC defense-linked institutions, underscoring the structural risk embedded within this program.[91]

90 See Journal Publication, Xiangyu Xu et al, Learning to Super-Resolve Blurry Face and Text Images, https://faculty.ucmerced.edu/mhyang/papers/iccv2017_gan_super_deblur.pdf, accessed on November 23, 2025
91 See Press Release, The Select Committee on the Chinese Communist Party, Moolenaar Touts Closure of Joint Programs Tied to CCP Tech Transfer, September 24, 2025, https://selectcommitteeontheccp.house.gov/media/press-releases/moolenaar-touts-closure-of-joint-programs-tied-to-ccp-tech-transfer

Notable Chinese Students and Scholars Supporting IARPA Research

Dingqiang Ye is a visiting scholar in the Department of Computer Science and Engineering at Michigan State University (inbound PhD student to Johns Hopkins University). He is also a master's student in the Department of Computer Science and Engineering at **Southern University of Science and Technology** in China. His research interests include computer vision, with a focus on pedestrian analysis, gait recognition, and representation learning. Ye previously was a Software Engineer intern at **Tencent** in 2021.[92] Ye's expertise combined with experience at Tencent, a company deeply integrated into PRC surveillance initiatives, positions him to contribute directly to China's next-generation biometric tracking and public security AI systems upon reintegration. Tencent is well-documented to facilitate censorship, mass surveillance, state propaganda, and human rights abuses on behalf of the Chinese Communist Party.[93]

Jie Zhu is currently a second year Ph.D. student in the Department of Computer Science and Engineering at Michigan State University in which he started Aug 2023 and is planned to be completed in April 2028. Zhu received his bachelors in June 2020 from **Northeastern University in China** and was an intern at **Seeking AI Co. Ltd in Guangzhou China** from December 2019 to April 2020, then was an **AI Engineer at the Tsinghua Pearl River Delta Research Institute (清华珠三角研究院)** from September 2020 to August 2021.[94] His research interests include computer vision, with a focus on multi-modal, fine-grained understanding, and representation learning.

- Zhu has several Chinese patents that coincide with his time in the United States and are counter to his public CV which states he was affiliated with the **Tsinghua Pearl River Delta Research Institut**e until April 2020. **The Guangdong Provincial People's Government and Tsinghua University** signed an agreement to jointly establish the Tsinghua Pearl River Delta Research Institute.[95]
- During Zhu's time in the U.S., even while supporting IARPA research, he has filed numerous patents in China assigned to **Cyagen (Guangzhou) Biosciences Co., Ltd. (赛业（广州）生物科技有限公司) and Tsinghua Pearl River Delta Research Institute**. The patents include:
    - CN113194348B, "Virtual human lecture video generation method, system, device and storage medium", granted: July 2022.
    - CN112562720B, "Lip-sync video generation method, device, equipment and storage medium", granted: July 2024.

92 See https://bugjudger.github.io/, date accessed September 3, 2025
93 See https://www.hrw.org/news/2020/09/28/chinese-tech-firms-fueling-beijings-repression, date accessed November 23, 2025
94 See https://jiezhu23.github.io/files/Resume-Jie%20Zhu.pdf, date accessed September 3, 2025
95 See http://www.tsinghua-gd.org/about.aspx, date accessed September 3, 2025

- CN113192161B, “Virtual human image video generation method, system, device and storage medium”, granted: October 2022.
- CN113192162B, “Method, system, device and storage medium for driving image by voice”, granted: December 2022.
- CN112487978B, “A Method for Speaker Localization in Video”, granted: April 2024.
- CN112562721B, “Method and device for positioning speaker in video and computer storage medium”, granted: April 2024.
- CN113179449B, “Method, system, device and storage medium for driving image by voice and motion”, granted: April 2022.
- CN112562719B, “Method, system, device and storage medium for matching synthesized voice with original video”, granted: March 2024.
- CN112530401B, “Speech synthesis method, system and device”, granted: May 2024.
- CN112565885B, “Video segmentation method, system, device and storage medium”, granted: January 2023.
- CN112530400A, “Method, system, device and medium for generating voice based on text of deep learning”, filed: November 2020 (pending).
- CN113259778A, “Method, system and storage medium for using virtual character for automatic video production”, filed: April 2021 (pending).

Zhu’s ongoing Chinese patent activity in virtual human modeling, speech synthesis, and video analytics filed while working on IARPA-funded projects in the U.S., demonstrates active technology transfer potential, with applications well-suited for PRC surveillance, influence operations, and synthetic media tools favored by MSS and MPS.

Chair Professor Kaifeng Jiang, currently at Peking University in Beijing, was previously a full professor at The Ohio State University’s Translational Data Analytics Institute. He supported a $7.9 million IARPA program focused on unobtrusive, multimodal sensing to model workforce performance. Now he supports the Guanghua Thinking Power Research Project (CNY 200,000) and has participated in one National Natural Science Foundation of China key project and four general projects.[96,97]

96 See https://en.gsm.pku.edu.cn/faculty/jiang.1788/, date accessed September 3, 2025
97 See https://www.kaifengjiangshrm.com/_files/ugd/1d996e_eadf996a4aa749d4923998c66bdfa03e.pdf, date accessed September 3, 2025

Dr. Yanmin Qian is a Full Professor at Shanghai Jiao Tong University and a prominent figure in speech and language processing. His work is deeply embedded in China's state research apparatus, including major projects funded by the Ministry of Science and Technology, the National Natural Science Foundation of China, and defense-linked actors such as AVIC. He previously supported the U.S. IARPA Babel program on multilingual speech recognition. At the same time, he contributed to **PRC defense and Ministry of National Defense projects**. This dual track places him squarely at the nexus of U.S.-funded intelligence research and China's military-aligned technology development.[98,99]

Dr. Ruonan Liu exemplifies the dual-track risks of U.S.–China research entanglement. While a joint postdoctoral fellow at Carnegie Mellon University, she supported the IARPA-funded DIVA program (contract D17PC00340) focused on deep intermodal video analytics, a project with clear implications for U.S. national security and intelligence applications. Upon returning to China, she transitioned into PRC-funded defense research, including a **National Defense Basic Scientific Research Program key project on "XX system XX positioning, diagnosis, and prediction,"**[100] **indicating involvement in sensitive or classified military technologies**. This trajectory underscores how expertise gained through U.S. intelligence-sponsored programs can be redirected toward advancing China's defense capabilities.[101,102]

Lanqing Guo is a postdoc research fellow at The University of Texas at Austin. She earned her Ph.D. in Electrical and Electronic Engineering from Nanyang Technological University, Singapore, where she graduated with the Best Thesis Award. Guo received her bachelor's from **Wuhan University**. Prior to going to Singapore, she was at the **AI Lab at Tencent, DAMO Academy at Alibaba, and the Multimedia Lab at Shenzhen Institutes of Advanced Technology (SIAT)**.[103] Her research interests include 2D/3D image processing and generation, computational imaging, and computer vision.[104] Guo's background in 2D/3D image processing, computational imaging, and prior work with Tencent, Alibaba DAMO Academy, and SIAT gives

98 See https://audiocc.sjtu.edu.cn/en/members/yanmin.qian, date accessed September 3, 2025
99 See https://audiocc.sjtu.edu.cn/zh/members/yanmin.qian/resume.pdf, date accessed September 3, 2025
100 Chinese researchers routinely use "XXX" in their publications to significate the research was classified or supporting weapons equipment development.
101 See https://automation.sjtu.edu.cn/Ruonan, date accessed September 4, 2025
102 See https://sais.sjtu.edu.cn/faculty/liuruonan.html, date accessed September 4, 2025
103 See https://guolanqing.github.io/, date accessed September 3, 2025
104 See https://ieeexplore.ieee.org/author/37088934155, date accessed September 3, 2025

her direct applicability to PRC state security imaging programs, including surveillance and AI-enhanced reconnaissance.

- Alibaba's DAMO Academy is explicitly named in official Hangzhou Municipal Government planning documents as a contributor to China's military-civil fusion (MCF) initiatives.[105] The same document also outlines a focus on key surveillance technologies, including biometrics, video and image data processing, video surveillance cloud retrieval, and panoramic super-view monitoring.[106] Alibaba DAMO has also been identified to conduct research in network refining for surveillance object detection, person re-identification for matching across camera surveillance systems, and facial recognition.[107] These technology areas are core enablers of state surveillance and public control systems, and internal security operations.

Yiyang Su is a Ph.D. student in the Department of Computer Science and Engineering at MSU. He received his B.S. degrees from the University of Rochester. He first came to the U.S. in 2017 after his time at **Tsinghua University, a SASTIND co-administered school**, in China (Where he learned about how to use neural networks to classify images).[108] His research interests include biometrics and image generation.[109] Su recently published the paper "SapiensID: Foundation for **Human Recognition**" In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.[110] Su's specialization in biometrics and human recognition, reinforced by his training at Tsinghua University, aligns closely with PRC defense and intelligence priorities, making his skillset valuable for MSS-led counterintelligence and MPS surveillance network optimization.

Allowing PRC-linked students and scholars to train inside U.S. research environments, particularly on IARPA-funded programs, creates a direct pipeline that equips future contributors to China's military-civil fusion enterprise with early access to advanced technologies, hands-on experience with U.S. government systems, and a permissive ecosystem for unmonitored patenting and technology how-how transfers. Once placed inside U.S. labs or under U.S. academics to work on IARPA-funded research, they gain exposure to cutting-edge biometric recognition, speech processing, multimodal sensing, and synthetic media research, capabilities directly sought by the MSS, MPS, and PLA research units. In no circumstance should the United States normalize these

105 See Public Announcement, Hangzhou Municipal Government, Implementation Opinions of the General Office of the Hangzhou Municipal People's Government on Accelerating the Development of the Military-Civilian Integration Industry Hang Zheng Ban Han [2018] No. 146 杭州市人民政府办公厅关于加快军民融合产业发展的实施意见 杭政办函〔2018〕146号, December 13, 2018, http://hzldzy.com/detail-3961.html
106 Ibid.
107 See https://selectcommitteeontheccp.house.gov/sites/evo-subsites/selectcommitteeontheccp.house.gov/files/evo-media-document/fox-in-the-henhouse-report-compressed_0.pdf, date accessed November 23, 2025
108 See https://www.linkedin.com/in/yiyang-su/details/experience/, date accessed September 3, 2025
109 See https://cse.msu.edu/~suyiyan1/cv/, date accessed September 3, 2025
110 See https://cse.msu.edu/~suyiyan1/, date accessed September 3, 2025

relationships or legitimize the presence of individuals or institutions that act as conduits for PRC military and security services, especially on USIC-funded research programs.

Reintegration Risk into PLA, MPS, and MSS Surveillance and Intelligence Ecosystems

The IARPA-supported students and researchers profiled in this report are not operating in an academic vacuum. Each is historically affiliated with elite Chinese universities that house state-validated National Defense Key Laboratories (国防重点学科实验室), specialized research centers designated to support PLA, public security (MPS), and intelligence (MSS) modernization efforts. These laboratories are instrumental to advancing dual-use technologies such as biometric surveillance, high-resolution imaging, synthetic vision, trusted computing, and precision navigation, domains that mirror the technical scope of current IARPA-funded programs.[111]

These individuals, many of whom have trained under or collaborated on IARPA-funded projects to include facial recognition, gait analysis, body-shape identification, or multimodal sensing, are positioned to seamlessly transition into China's broader state security apparatus. Upon return to China, or through persistent remote collaboration, they may be vectored toward MSS bureaus, MPS surveillance divisions, or PLA-affiliated laboratories:[112,113]

Call to Action: Prohibit Adversarial Collaboration in Intelligence and Defense Research

The evidence presented in this report reveals a systemic failure to protect U.S. intelligence-funded research from foreign adversarial access. Despite growing recognition of the threat posed by the Chinese Communist Party's military-civil fusion strategy, researchers from PRC institutions, including those directly affiliated with the PLA, nuclear weapons development, and state surveillance, have repeatedly participated in IARPA-funded programs and other sensitive U.S. government research initiatives.

The ODNI, IARPA, DARPA, and all defense-related funding agencies must be explicitly prohibited from engaging in any collaboration with institutions located in or affiliated with adversarial defense and security-affiliated entities. This includes joint authorship, collaboration,

111 See https://www.aspi.org.au/report/china-defence-universities-tracker/, date accessed September 3, 2025
112 See https://www.airuniversity.af.edu/LinkClick.aspx?fileticket=BeogKkr0Pcw%3D&portalid=10, date accessed September 3, 2025
113 See https://www.dataabyss.ai/platform-resources/prc-defense-laboratories, date accessed September 3, 2025

subcontracting, co-funding, or presenting at institutions tied to adversarial military, intelligence, or government agencies.

Safeguarding America's technological edge requires more than awareness, it requires enforcement. Congress and executive agencies must establish non-negotiable guardrails that firewall U.S. national security science from adversarial exploitation. The cost of inaction is the continued erosion of strategic advantage and the empowerment of foreign regimes through U.S. taxpayer-funded innovation.

Recommendations

1. Executive Branch agencies should update award terms and conditions to clearly prohibit any research relationship, formal or informal, on funded projects with entities already designated on U.S. government blacklists.
2. Congress should enact and codify clear prohibitions on using taxpayer funds to support research collaborations with defense and security entities affiliated with foreign adversarial nations.
3. As a condition of award, U.S. professors receiving IARPA-funded research should be restricted from traveling to adversarial countries for the duration of the project and for two years following its completion.
4. Foreign students from adversarial countries should be prohibited from participating in research conducted under IARPA-funded awards.
5. Conduct damage assessments on research identified in this report. This should include a review of affected grants to determine whether research outcomes transitioned into controlled programs (e.g., SBIR/STTR, classified projects, or special access programs) supporting warfighting capabilities, and whether related advances were concurrently or subsequently diverted to China's military through research collaboration. This assessment is critical to determining the extent of technology leakage into China's defense research and industrial base, whether adversaries now possess comparable capabilities or have already developed countermeasures.
6. The Executive Branch should establish a National Research Security, Integrity, and Compliance Center (NRSICC) and congress should codify it in legislation.
    a. The United States government and its federally funded research enterprise face persistent challenges in protecting critical and emerging technologies from exploitation by foreign adversaries. The current decentralized framework, spread across multiple federal funding agencies and institutions of higher education, has proven insufficient, inconsistent, and ineffective in preventing the diversion of taxpayer-funded research, intellectual capital, and technological know-how. To address this, the Executive Branch and Congress should authorize and fund the creation of the NRSICC. This new centralized entity would consolidate all federal

research security and due diligence efforts, establish uniform standards and policies, and provide cross-agency oversight and coordination. The NRSICC would operate as a centralized authority responsible for risk assessments, compliance monitoring, investigative lead generation, and engagement with both academia and industry. Its creation would eliminate duplicative efforts, close enforcement gaps, reduce costs, and significantly enhance the federal government's ability to protect U.S. technological and economic leadership.

i. Current research security efforts are fragmented across federal agencies, with varying standards, limited data sharing, and inconsistent due diligence protocols.
ii. Duplication of effort and lack of automation result in wasteful spending, slow assessments, and gaps in monitoring and compliance enforcement.
iii. Foreign adversaries, particularly the PRC, exploit these weaknesses, targeting U.S. research infrastructure through talent programs, joint research partnerships, and statutory mechanisms that mandate acquisition of foreign science and technology.
iv. The NRSICC will centralize policy, oversight, and technical solutions, enabling the federal government to more effectively assess risk, detect compliance failures, and protect national security interests in the research ecosystem.

Policy Development and Oversight:

- Establish and enforce uniform federal standards for research security and due diligence across fundamental research and SBIR/STTR programs.
- Standardize proposal forms, reporting documents (e.g., RPPR), and risk assessment templates across all agencies.
- Maintain and update a consolidated list of prohibited or high-risk foreign entities, including malign talent programs, defense-affiliated institutions, and sanctioned organizations.
- Serve as the coordinating body for research security policy across the federal government, including alignment with export control and national security directives.

Due Diligence and Risk Assessments:

- Conduct all pre-award due diligence and post-award compliance monitoring for fundamental research and SBIR/STTR contracts.
- Use automated tools and data aggregation to streamline initial risk flagging and enable deep-dive human analysis where needed.
- Implement a standardized, tiered risk assessment framework to guide funding decisions and mitigation strategies based on quantitative and qualitative indicators.

- Conduct ongoing monitoring of research activities and publications to detect noncompliance or emerging risk.

Investigative Coordination and Compliance:

- Refer criminal, civil, or administrative violations identified during the due diligence process to appropriate law enforcement or oversight bodies (e.g., DOJ, OIG, DCIS).
- Support investigative task forces and provide technical expertise in research security cases.
- Oversee and manage a Voluntary Disclosure Program for institutions of higher education, offering reduced penalties for past compliance failures in exchange for proactive identification and resolution of those failures.

Technology Infrastructure and Automation:

- Build and maintain a centralized IT infrastructure capable of aggregating relevant government and public data (e.g., visa data, travel records, patents, Section 117 disclosures, federal grant records).
- Integrate and automate entity screening using machine-readable forms and customizable filters based on evolving risk indicators.
- Maintain an integrated list combining the NDAA Section 1286 List, BIS Entity List, OFAC sanctions, and other risk designations.

Public-Private Sector Engagement:

- Serve as the primary federal interface on research security for academia, industry, and allied governments.
- Promote transparency, compliance best practices, and two-way communication with research institutions and federal award recipients.
- Serves as the lead government entity to share research security information and training materials.

The NRSICC's Estimated Cost and Funding Model:

- Implement a 0.1% to 0.4% "Research Security Due Diligence Fee" on federal fundamental research and SBIR/STTR awards. This is modeled after the existing "Facilities and Administrative" (F&A) rate and would fund all NRSICC operations without requiring additional appropriations.
- Consolidating existing contracts, licensing, and due-diligence functions across agencies, while repurposing the NSF SECURE Center and integrating its capabilities into the NRSICC, is projected to yield $10–$20 million in savings over several years, with additional multi-million-dollar cost avoidance.

APPENDIX: Additional IARPA Case Studies

Case Study 1: December 2024 – *Pyramid Diffusion for Fine 3D Large Scene Generation* This IARPA-funded project (contract 140D0423C0074) advances 3D large-scene generation using pyramid diffusion models, a cutting-edge technique with direct applications in surveillance, geospatial intelligence, and autonomous system navigation. The research was co-authored by personnel from the University of California-Merced, NVIDIA, City University of Hong Kong, Yonsei University, and the University of Leeds.[114] Critically, it also involved Yuheng Liu and Chongshou Li of **Southwest Jiaotong University**, a PRC institution known for AI and transportation systems research with military relevance.[115] [116] The authors were also funded by the National Natural Science Foundation of China (NSFC Grant 62202395), creating a clear dual-funding entanglement between IARPA and PRC state-backed science funding (NSFC). The participation of PRC-based researchers in a project of this nature poses substantial national security concerns, particularly due to the dual-use potential of the technology in battlefield mapping, smart surveillance, and autonomous vehicle targeting systems.[117]

Official implementation of "Pyramid Diffusion for Fine 3D Large Scene Generation"

Case Study 2: ***"Attentional Ptycho-Tomography for 3D Nanoscale Imaging"***[118] (2023) funded by IARPA contract FA8650-17-C-9113, which included **ShanghaiTech University**, an institution under the Chinese Academy of Sciences. The technology, capable of nanoscale 3D imaging with minimal data, has direct implications for semiconductor inspection, materials espionage, and stealth analysis.

Case Study 3: ***"ImU: Physical Impersonating Attack for Face Recognition System with Natural Style Changes"***[119] (2023) and ***"OSPREY: Recovery of Variable and Data Structure via Probabilistic Analysis***

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114 See https://github.com/yuhengliu02/pyramid-discrete-diffusion, date accessed September 3, 2025
115 See https://www.swjtu.edu.cn/kxyj1.htm, date accessed November 24, 2025
116 See https://web.archive.org/web/20250912082253/https://www.scmp.com/news/china/military/article/3275645/pla-rocket-force-bans-3-universities-procurement-activities-over-bid-rigging, date accessed November 24, 2025
117 Liu, Y., Li, X., Li, X., Qi, L., Li, C., Yang, MH. (2025). Pyramid Diffusion for Fine 3D Large Scene Generation. In: Leonardis, A., Ricci, E., Roth, S., Russakovsky, O., Sattler, T., Varol, G. (eds) Computer Vision – ECCV 2024. ECCV 2024. Lecture Notes in Computer Science, vol 15127. Springer, Cham. https://doi.org/10.1007/978-3-031-72890-7_5
118 Kang, I., Deng, M., Ge, B., et al. (2023). Attentional Ptycho-Tomography (APT) for three-dimensional nanoscale X-ray imaging with minimal data acquisition and computation time. Light: Science & Applications, 12, 103. https://doi.org/10.1038/s41377-023-01181-8
119 Li, X., Zhao, W., Zhou, Y., et al. (2023). ImU: Physical impersonating attack for face recognition system with natural style changes. 2023 IEEE Symposium on Security and Privacy (SP). https://doi.org/10.1109/sp46215.2023.10179360

for Stripped Binary"[120] (2021) funded by IARPA TrojAI Grant W911NF-19-S-0012, and Office of Naval Research Grants N000141712045, N000141410468 and N000141712947 involved researchers from **Nanjing University and Renmin University of China**, respectively. These two studies focused on biometric spoofing and binary reverse engineering have clear offensive and defensive implications for cyber operations and surveillance evasion. "OSPREY" additionally involved Chinese co-funding from National Natural Science Foundation of China and Fundamental Research Funds for the Central Universities and the Research Funds of Renmin University of China.

Case Study 4: ***"CRISPR Screens in Drosophila Cells Identify Vsg as a Tc Toxin Receptor"***[121] (2022) funded by IARPA Grant W911NF-17-2-0089 included a researcher from **Xuzhou Medical University**, working on toxin-gene interactions that could be repurposed for biological warfare or synthetic neuromodulation. ***"Targeted Intracellular Delivery of Cas13 and Cas9 Nucleases Using Bacterial Toxin-Based Platforms"***[122] (2022) funded by IARPA via Army Research Office Grant W911NF-17-2-0089, involving a researcher **from The First Hospital of Jilin University**, focuses on precision gene-editing delivery mechanisms using bacterial toxin pathways to transport CRISPR-based tools directly into living cells. This technique has implications for therapeutic engineering, but also for synthetic bioweapon delivery, neuromodulation, and immune system manipulation.

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120 Z. Zhang et al., "OSPREY: Recovery of Variable and Data Structure via Probabilistic Analysis for Stripped Binary," 2021 IEEE Symposium on Security and Privacy (SP), San Francisco, CA, USA, 2021, pp. 813-832, doi: 10.1109/SP40001.2021.00051
121 Zhang, Y., Kim, J., Chen, H., et al. (2022). CRISPR screens in Drosophila cells identify Vsg as a Tc toxin receptor. Nature, 609(7928), 333–339. https://doi.org/10.1038/s41586-022-05250-7
122 Anderson, E., Chen, M., Sheng, K., et al. (2022). Targeted intracellular delivery of Cas13 and Cas9 nucleases using bacterial toxin-based platforms. Cell Reports, 38(9), 110476. https://doi.org/10.1016/j.celrep.2022.110476

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John Sava is the Senior Investigator for the Congressional Select Committee on China, where he leads investigations and legislative initiatives focused on research security, foreign influence, technology transfer, and intellectual property theft. He is widely recognized within the US intelligence and law enforcement communities as a subject-matter expert on the People's Republic of China's (PRC's) technology transfer apparatus and defense research and industrial base. Mr. Sava previously served as a special agent/criminal investigator within the Department of Defense, where he conducted counterintelligence investigations and operations. Over the course of his career, Mr. Sava has been called upon as an expert witness and briefer on PRC-related issues to numerous congressional committees. From 2019 to 2025, Mr. Sava served as Co-Chair of the Foreign Influence Investigations Working Group for Inspectors General and law enforcement across the US Government. He is currently a lecturer on China, research security and due diligence, and technology protection at the Department of Defense Joint Counterintelligence Training Academy, private industry, and institutions of higher education. Mr. Sava has received numerous national awards, including the National Counterintelligence and Security Center Counterintelligence Analysis Award (2021).